

3/23

Windows server installed

3/24

Opnsense installed

Win11 installed

Ubuntu installed

Joined win11 to AD

3/28-3/31

Vlans

AD is on a vlan

IP: 192.168.10.2

Only certain ports are allowed

Dns points to itself but forwards to 1.1.1.1

Firewall: Rules: AD_VLAN10									
53 88 135 389 445 464 636 3268:3269 49152:65535 Inspect									
☐	Protocol	Source	Port	Destination	Port	Gateway	Schedule		Description ?
☐									Automatically generated rules
☐	IPv4 → TCP/UDP 	*	*	AD_Server_IP	AD_Ports	*	*		
☐	IPv4 → TCP/UDP 	AD_Server_IP	*	1.1.1.1/32	DOMAIN (53)	*	*		allows ad to forward dns out
☐	IPv4 →	*	*	*	*	*	*		
☐	IPv4 ←	*	*	*	*	*	*		

Employees are on a vlan

Employees do not have management acces (ability to ssh or use firewall webpage)

Employees can not communicate with other vlans

AD, and the web server are the only IPs outside the employee vlan they can talk to



OPNsense®
Securing networks made easy

root@OPNsense.internal

NAT

Rules [new]

Rules

Migration assistant

Floating

Admin_VLAN1

AD_VLAN10

BYOD_Vlan50

Employee_VLAN30

guest_VLAN60

IOT_VLAN40

WAN

Web_VLAN20

Shaper

Settings

Log Files

Diagnostics

VPN

Services

Power

IPv4

TCP/UDP

net

AD_Server_IP

*

*

*

IPv4

*

*

*

WEB_IP

*

*

*

IPv4

*

Employee_VLAN30

*

Admin_VLAN1 net , Web_VLAN20 net , AD_VLAN10 net , BYOD_Vlan50 net , guest_VLAN60 net , [...]

*

*

*

IPv4

TCP/UDP

net

This Firewall , Employee_VLAN30

admin_controls

*

*

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BYOD,IOT,and guest vlans
are not able to communicate with anything internal

Automatic generated rules									
☐		IPv4 *	IOT_VLAN40 net	*	Admin_VLAN1 net , Employee_VLAN30 net , Web_VLAN20 net , AD_VLAN10 net , BYOD_Vlan50 net , [...]	*	*		
☐		IPv4 TCP/UDP	IOT_VLAN40 net	*	IOT_VLAN40 address	admin_controls	*	*	disables management interfaces
☐		IPv4 *	*	*	*	*	*	*	
☐		IPv4 *	*	*	*	*	*	*	
pass block reject log in first match pass (disabled) block (disabled) reject (disabled) log (disabled) out last match									

Web server is on a vlan

IP: 192.168.20.3

Web server dns points to vlan ip as to not talk to internal systems

Only internal things able to communicate with web server are admin and employee vlans, (employee because its not a public website, else employees could reach it from the internet instead of internal)

☐	Protocol	Source	Port	Destination	ports allowed	Gateway	Schedule	Description
☐					53 80 123 443			Automatically generated rules
☐	IPv4 → TCP/UDP  	*	*	WEB_IP	WEB_traffic	*	*	
☐	IPv4 → TCP/UDP  	Web_VLAN20 net	*	Admin_VLAN1 net , Employee_VLAN30 net , Web_VLAN20 net , AD_VLAN10 net , BYOD_Vlan50 net , [...]	*	*	*	blocks web server from
☐	IPv4 → TCP/UDP  	Web_VLAN20 net	*	Web_VLAN20 address	admin_controls	*	*	disables management interfaces
☐	IPv4 → TCP/UDP  	*	*	*	*	*	*	

Saved a snapshot after completion

Backups

Active Directory

Group policy

- Password policy implemented 14+ char, 26 password remembered, complexity requirement, reset every 90 days.
- Restricted admin policy: disallowed restricted admin logon so accounts can only be used to authenticate.